Transformers

AI506 — Advanced Machine Learning

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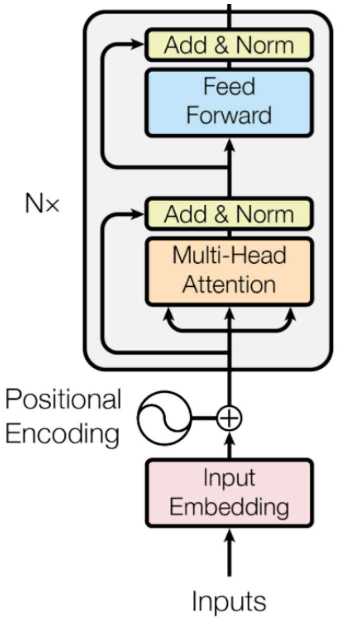
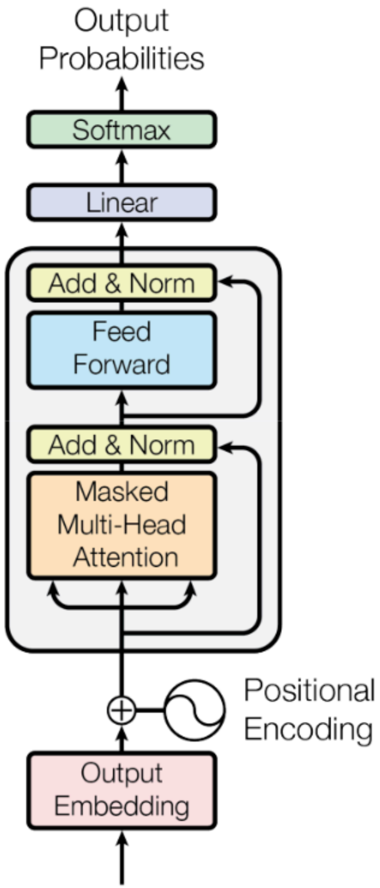
May 2026

The Transformer Architecture

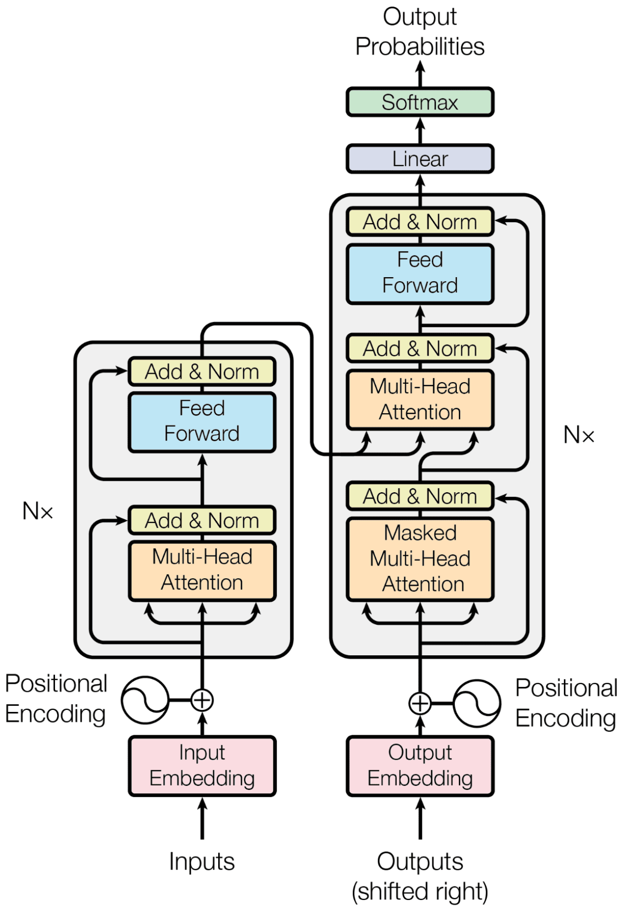
Encoder



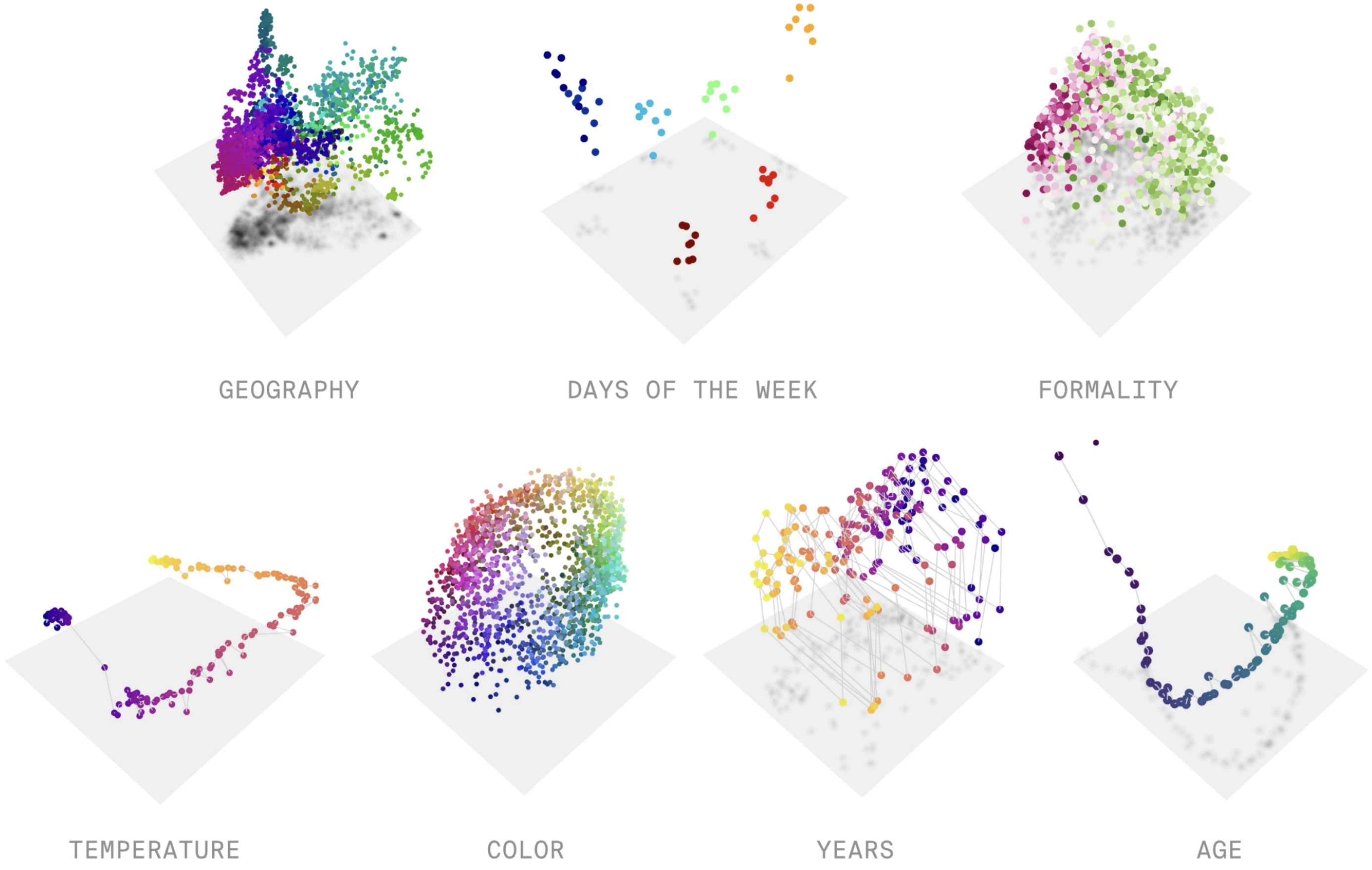
Decoder



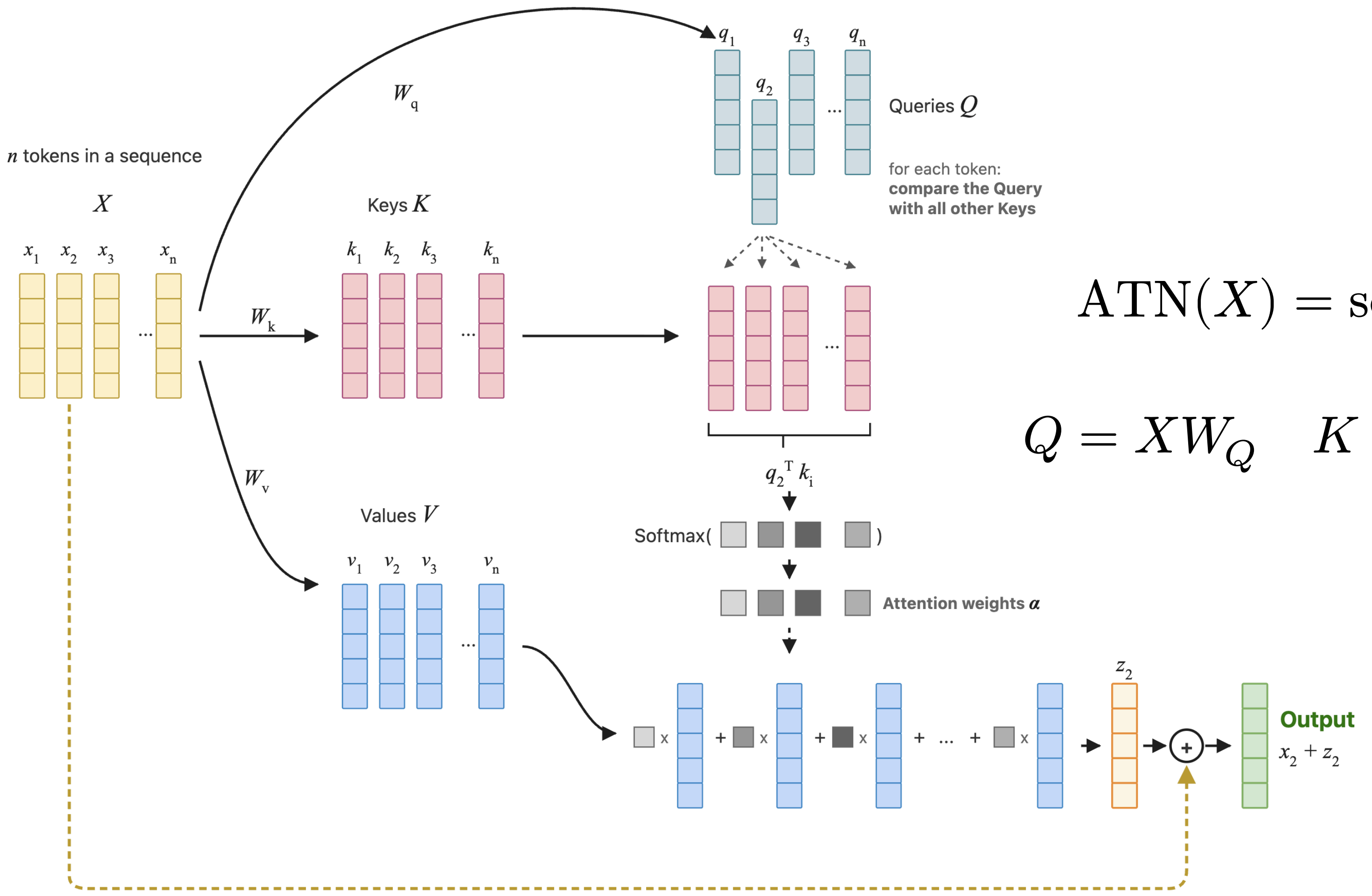
Encoder-Decoder



Tokenization and encoding



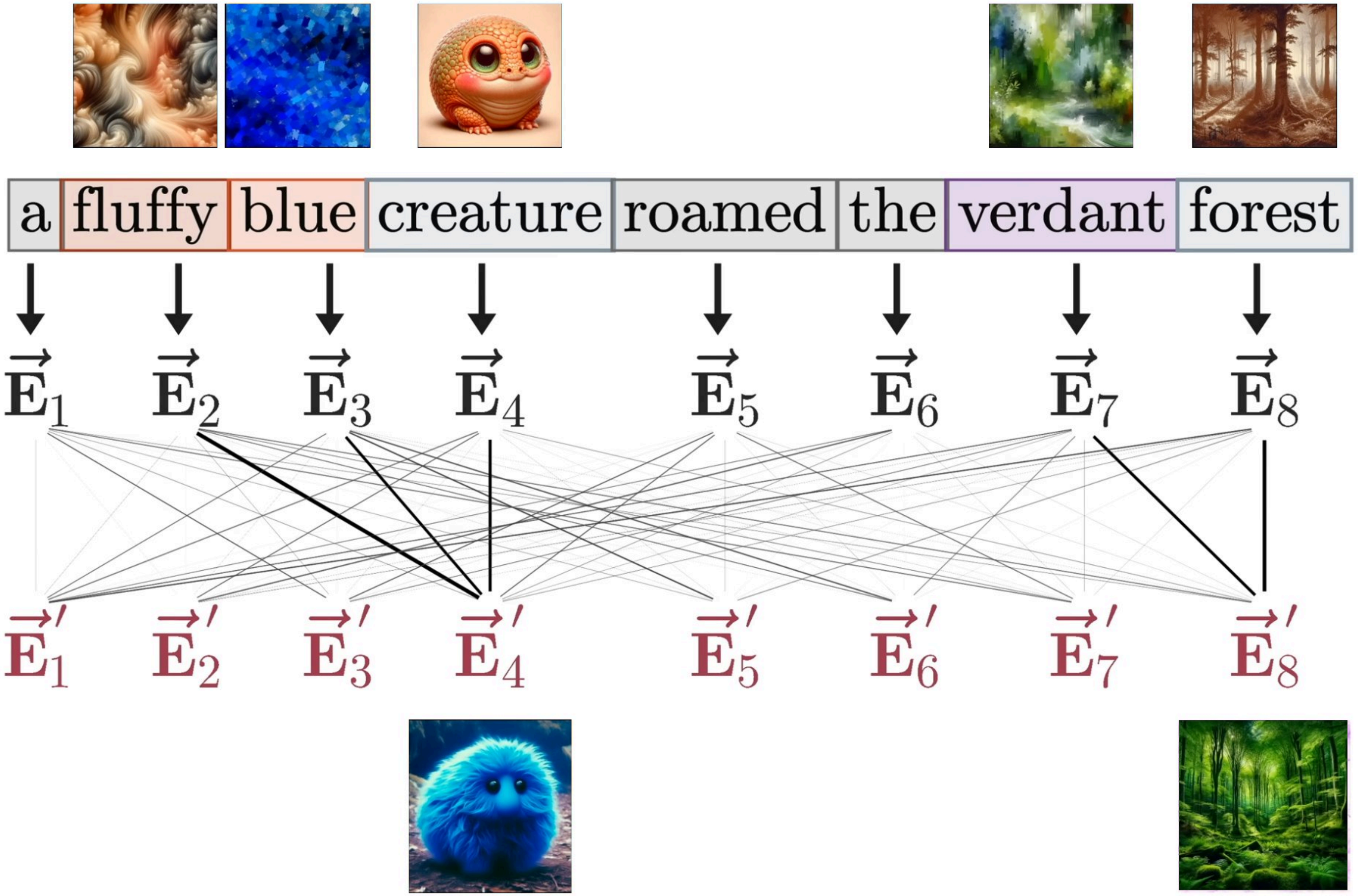
Self-attention



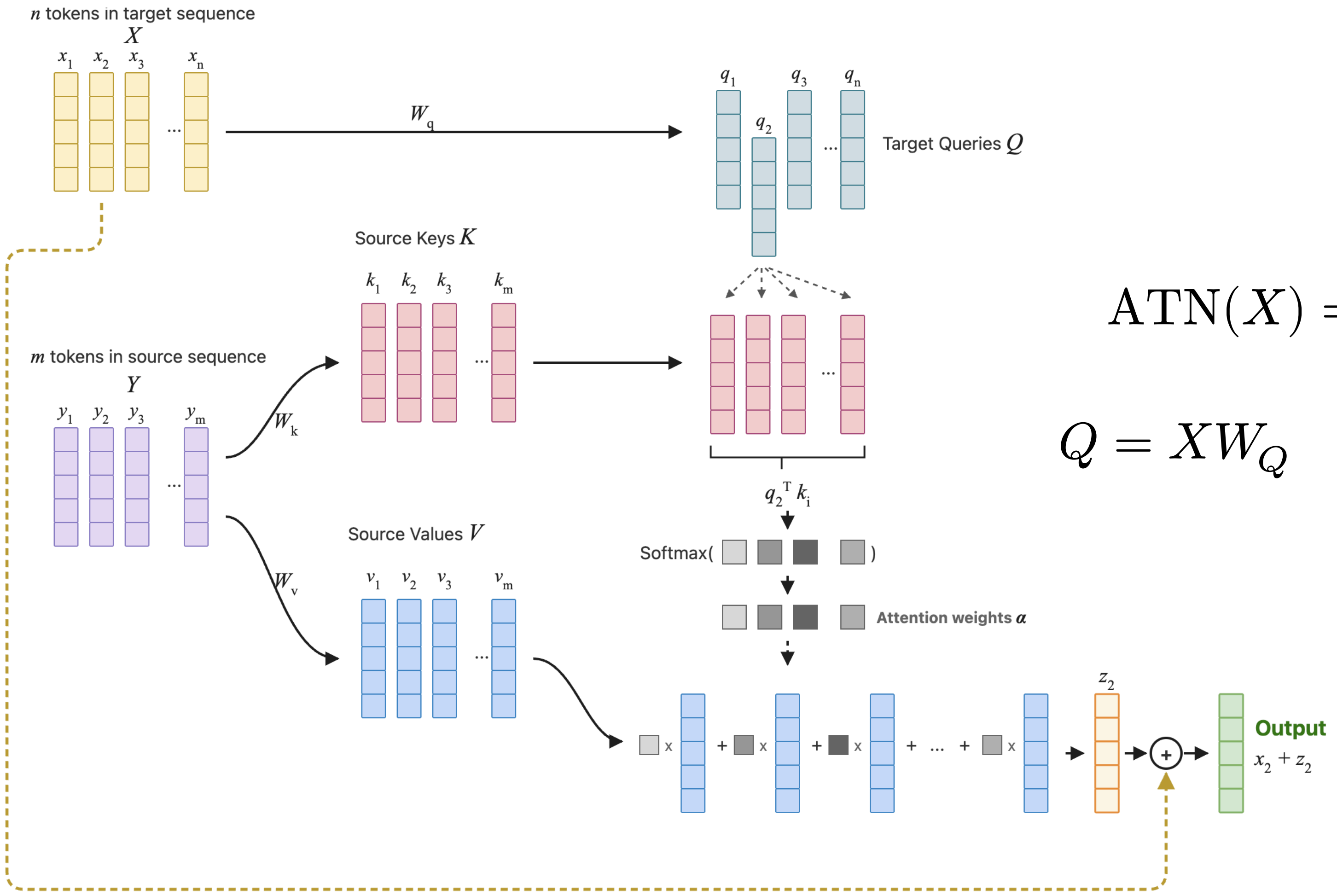
$$\mathrm{ATN}(X) = \mathrm{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

$$Q = XW_Q \quad K = XW_K \quad V = XW_V$$

Self-attention



Cross-attention



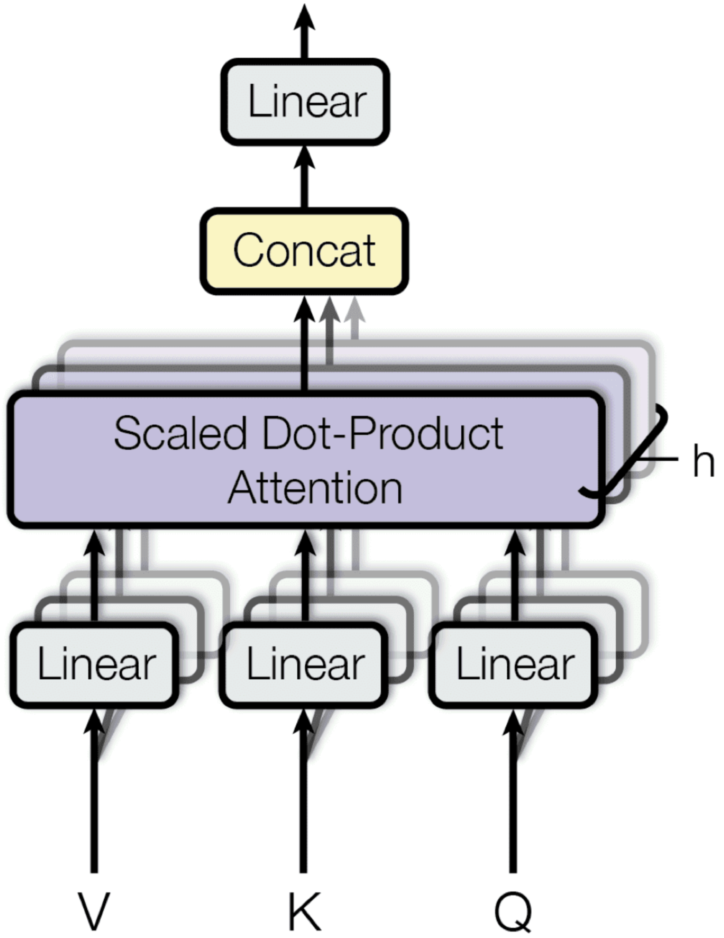
$$\mathrm{ATN}(X) = \mathrm{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

$$Q = XW_Q \quad K = ZW_K \quad V = ZW_V$$

Multi-head attention

$$\mathrm{MultiHead}(X) = \mathrm{head}_1 \mathbin{+\!\!+} \ldots \mathbin{+\!\!+} \mathrm{head}_h$$

$$\mathrm{head}_i = \mathrm{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$



Positional encoding

- Absolute position

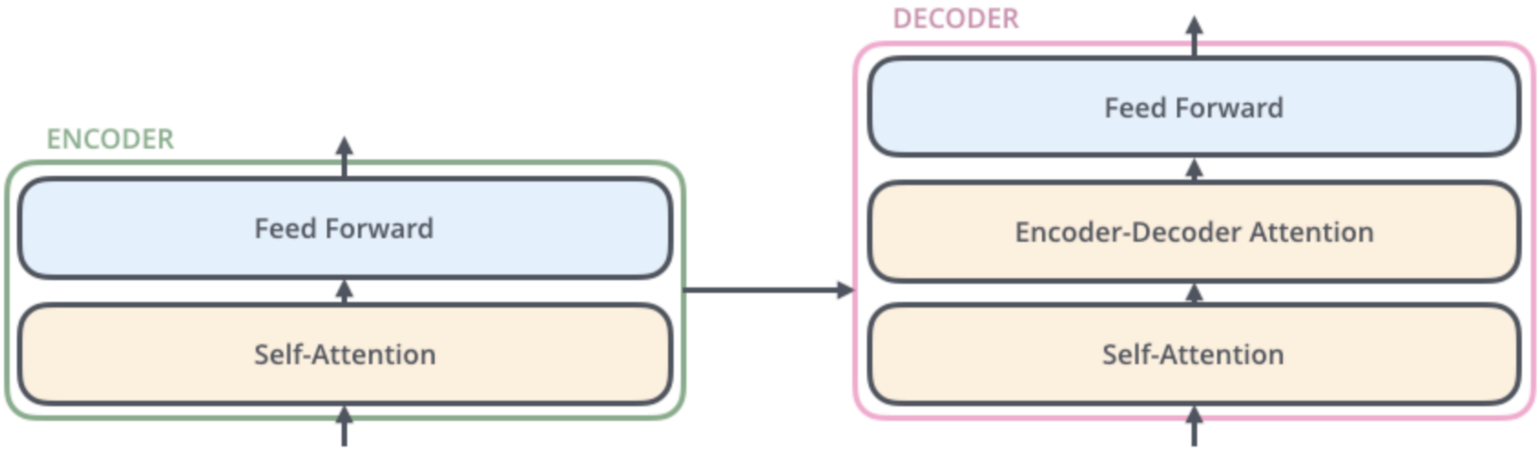
$$v_i = W_v \cdot (x_i + p_i)$$

- Relative position

$$q_m^T k_n = x_m^T W_Q^T W_K x_n + \boldsymbol{b_{n-m}}$$

Attention masks

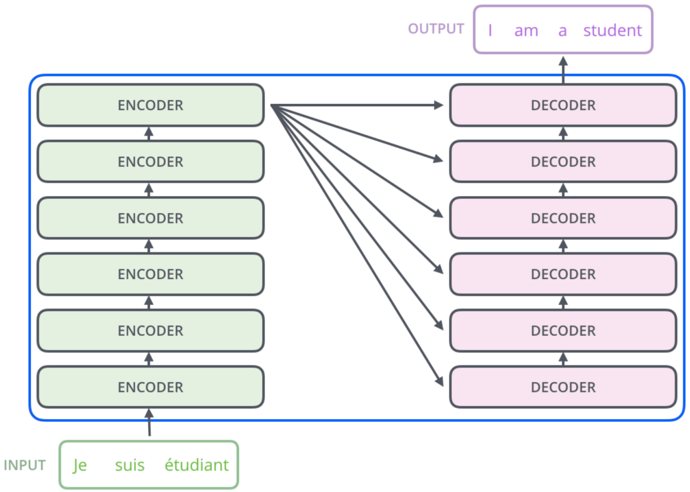
Pointwise MLP

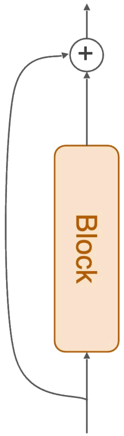


$$\text{MLP}(x) = W_{\text{out}} \cdot \sigma(W_{\text{in}} x + b_{\text{in}}) + b_{\text{out}}$$

Given this input, which learned patterns are relevant, and what should be added to the representation as a result?

Residual connections and Normalization





$$h^{(l+1)} = \mathrm{LN}\left(\mathrm{MLP}\left(h^{(l)}\right) + h^{(l)}\right)$$

$$h^{(l+1)} = \mathrm{MLP}\left(\mathrm{FN}\left(h^{(l)}\right)\right) + h^{(l)}$$

$$\mathrm{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$

Language modeling objective

- Pretraining

$$\mathcal{L}_{\mathrm{PT}} = -\sum_t \log p_\theta(x_t \mid x_{<t})$$

- Supervised Finetuning

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_t \log p_\theta(y_t \mid x, y_{<t})$$

- RL from Human Feedback